

CLASSIFICATION OF MILITARY RADIO FREQUENCY BANDS

MARINE CORPS COMMUNICATION ELECTRONICS SCHOOL (MCCES)

LESSON 1 – OPERATIONAL FOUNDATIONS

Based on MCRP 3-40.3C: Multi-Service Tactics, Techniques, and
Procedures for the Employment of Tactical Radios



SITUATION: THE COST OF COMMUNICATION FAILURE

Imagine you are in a high-intensity combat environment. You receive a tasking to establish a long-range link back to the Rear Operations Center.

THE PROBLEM

You are handed a list of frequencies. If you cannot distinguish between HF, VHF, and UHF, you will select:

- The wrong radio system
- The wrong antenna
- Result: Failure to establish communications.

GOAL

Understanding the spectrum is the first step in ensuring the MAGTF maintains its command and control edge.



MISSION: LEARNING OBJECTIVES

TERMINAL LEARNING OBJECTIVE (TLO-001)

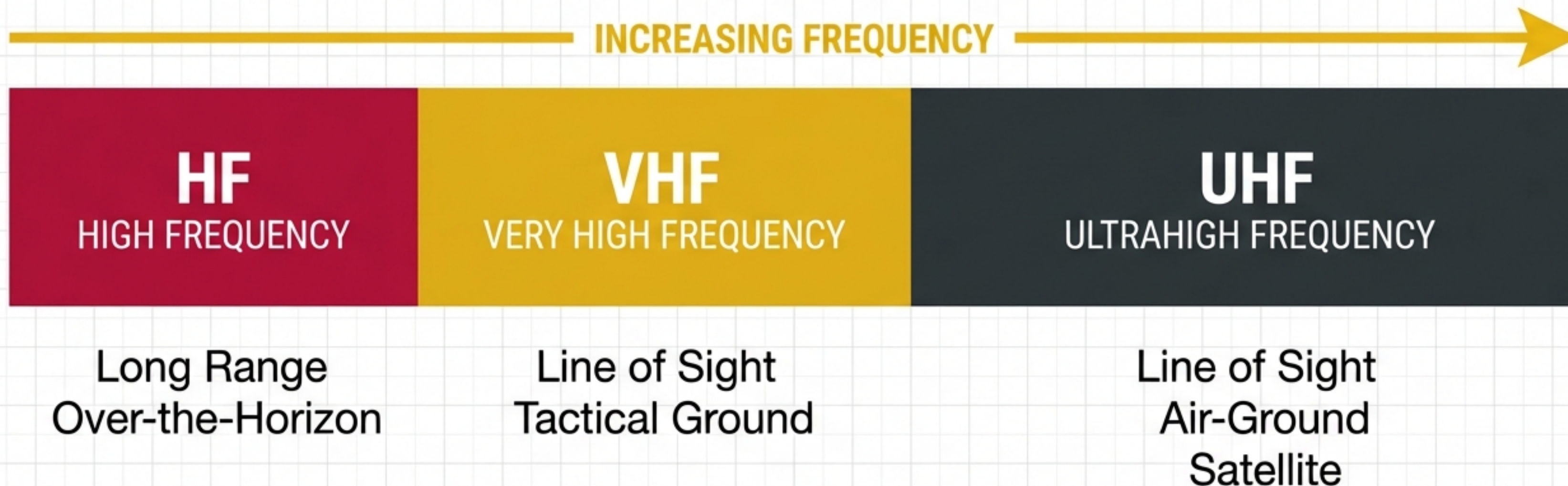
Given a list of tactical frequencies and MCRP 3-40.3C,
CLASSIFY each frequency into the correct military radio frequency band (HF, VHF, or UHF) without error.

ENABLING OBJECTIVES (TACTICAL TASKS)

1. **DEFINE** the frequency limits for HF, VHF, and UHF bands per Figure 1-1.
2. **IDENTIFY** the correct unit of measurement and convert all frequencies to a common unit (MHz).

INTEL: THE MILITARY RADIO FREQUENCY SPECTRUM

Per MCRP 3-40.3C, the tactical spectrum is divided into three primary bands.



INTEL: FREQUENCY BAND LIMITS

HF (HIGH FREQUENCY)	3 MHz to 30 MHz
VHF (VERY HIGH FREQUENCY)	30 MHz to 300 MHz
UHF (ULTRAHIGH FREQUENCY)	300 MHz to 3 GHz (3,000 MHz)

TEACHING POINT:

These limits are absolute per Figure 1-1. Memorize the boundaries: 3, 30, and 300.

TTPs: UNITS OF MEASUREMENT

Frequencies are expressed in Hertz (Hz). To classify accurately, you must ensure you are comparing “apples to apples”.

THE OPERATIONAL STANDARD: MEGAHERTZ (MHz)

1 GIGAHERTZ (GHz)

= 1,000,000,000 Hz

1 MEGAHERTZ (MHz)

= 1,000,000 Hz

1 KILOHERTZ (kHz)

= 1,000 Hz

1,000 kHz = 1 MHz

1,000 MHz = 1 GHz

TTPs: CONVERSION PROCEDURES

Goal: Convert all incoming data to MHz.

RULE 1: SMALL TO MEDIUM
(kHz -> MHz)



DIVIDE by 1,000

3,000 kHz $\xrightarrow{\div 1000}$ **3 MHz**

RULE 2: LARGE TO MEDIUM
(GHz -> MHz)



MULTIPLY by 1,000

1.2 GHz $\xrightarrow{\times 1000}$ **1,200 MHz**

Move the decimal point 3 places.

EXECUTION: PRACTICAL APPLICATION SCENARIO

ROLE: Radio Chief, Distributed Platoon.

TASK: Categorize the provided frequency list to assign to correct radio nets (PRC-150, PRC-117G, or PRC-152).



STEP 1: STANDARDIZATION (CONVERT TO MHz)

Apply conversion rules to ensure all data is in Megahertz.

12.500 MHz ✓ No Change

155.225 MHz ✓ No Change

2400 kHz — $\div 1,000 \rightarrow$ 2.4 MHz

1.2 GHz — $\times 1,000 \rightarrow$ 1,200 MHz

STEP 2 & 3: COMPARE AND CLASSIFY

Compare the MHz values against MCRP 3-40.3C limits to label the bands.

VALUE (MHz)	LOGIC CHECK	CLASSIFICATION
12.5 MHz	Between 3 - 30 MHz	HF
155.225 MHz	Between 30 - 300 MHz	VHF
2.4 MHz	Below 3 MHz threshold	HF (Verify Capability)
1200 MHz	Between 300 - 3000 MHz	UHF

DEBRIEF: OPERATOR'S SUMMARY

This summary consolidates key frequency ranges and conversion rules for quick reference.

FREQUENCY SPECTRUM CHEAT SHEET

HF:	3 - 30 MHz
VHF:	30 - 300 MHz
UHF:	300 MHz - 3 GHz

CONVERSION RULES

kHz -> MHz: **DIVIDE** by 1,000

GHz -> MHz: **MULTIPLY** by 1,000

The background of the slide is a photograph of a communication site at sunset. Silhouettes of various antennas, towers, and buildings are visible against a sky with orange and blue hues. A large white rectangular box with a light gray grid pattern is centered on the image, containing the main text.

FINAL CHARGE

Mastery of the frequency spectrum ensures that you employ the right equipment for the right mission.

Keep the lines of communication open when they matter most.